

AW-FNO: An Adaptive approach for Fluid flow Super-resolution via Gated Wavelet-Fourier Learning

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1. Introduction

The study of Partial Differential Equations (PDEs) governs the physical dynamics of a system [1]. Numerical methods solve PDEs with accuracy, but are computationally expensive. Particularly, fluid dynamics simulation at high Reynolds numbers becomes challenging due to the complexity of turbulent interactions. The high computational demand for generating high-fidelity (HF) data through Direct Numerical Simulation (DNS) makes it impractical for real-world applications. As a result, the reconstruction of high-resolution (HR) flow fields from low-resolution (LR) data, called Super Resolution (SR), is a focus of research.

Turbulent flows transfer kinetic energy from coherent large-scale eddies to dissipative small-scale structures. Conventional Convolutional Neural Networks (CNNs) falter due to their spatially local impeding global coherence capture. Neural Operators (NOs) have recently been used for SR paradigms [2] as they learn resolution-invariant mappings between infinite-dimensional function spaces [3]. Fourier Neural Operators (FNOs) [3, 4], leveraging fast Fourier transforms (FFTs) in the Fourier domain, resolve global periodicities but still falter at local discontinuities and sharp gradients.

This work proposes the Augmented wavelet Fourier Neural Operator (AW-FNO), a synergistic architecture fusing the Fourier Neural Operator (FNO) [4] with the Wavelet Neural Operator (WNO) [5]. It combines FNO's spectral efficiency for non-local correlations with WNO's multi-resolution wavelet decomposition. A parallel kernel integration allows hierarchical feature extraction across scales, resulting in superior turbulent flow super-resolution with enhanced generalisation and computational efficiency.

2. Methodology

2.1 The Architecture

We aim to learn an operator $\mathcal{G} : \mathcal{A} \rightarrow \mathcal{U}$ that maps a low-resolution input field $a \in \mathcal{A}$ to a high-resolution output field $u \in \mathcal{U}$.

The core of our approach is the parallel spectral-wavelet layer. The input features are passed through two concurrent paths. The Fourier branch performs a Fast Fourier Transform (FFT), applies a learnable weight matrix to the low-frequency modes, and truncates high-frequency noise. This captures a global structure of the flow. The wavelet branch applies a Discrete Wavelet Transform (DWT) to decompose the signal into different scales. This helps the model learn localised refinements, such as eddies and boundary layer fluctuations.

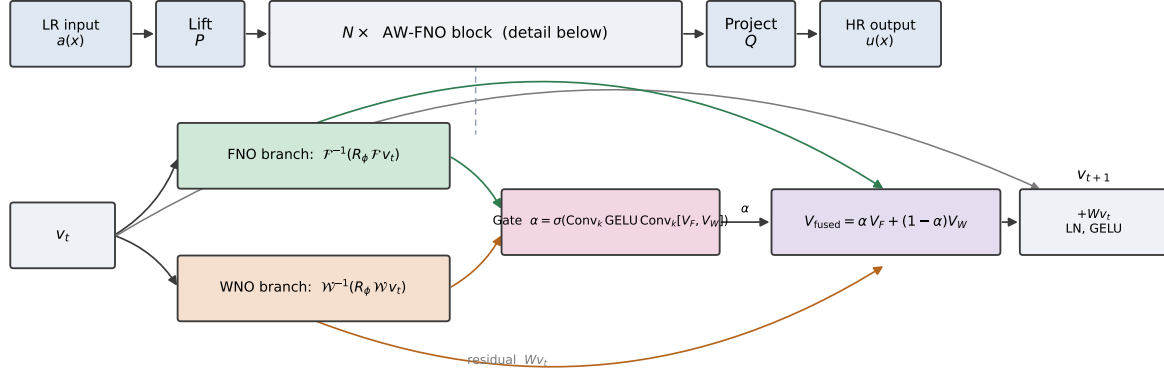


Figure 1: Architecture of the Augmented Wavelet Fourier Neural Operator

2.2 Mathematical Formulation

The update for a hidden state v_t to v_{t+1} fuses the two branches through a spatially varying gate $\alpha(x)$:

$$v_{t+1}(x) = \sigma \left(Wv_t(x) + \alpha(x) \odot (\mathcal{K}_F v_t)(x) + (1 - \alpha(x)) \odot (\mathcal{K}_W v_t)(x) + b(x) \right), \quad (1)$$

where $\alpha(x) \in [0, 1]$ is the adaptive gate produced by the gated fusion mechanism of Section 2.3, $\odot$ denotes element-wise (channel-wise) multiplication, x is input from the domain, W is a local linear Operator, $\mathcal{K}_F$ is the Fourier kernel operator, defined as $(\mathcal{K}_F(\phi) * (v_t))(x) = \mathcal{F}^{-1}(R_{\phi} \cdot \mathcal{F}(v_t))(x)$, where $\mathcal{F}$ is Fourier transform, $\mathcal{F}^{-1}$ is inverse Fourier transform, R_{ϕ} is a neural network-parameterized representation of the kernel k in Fourier space with some parameter $\phi \in \mathcal{R}^P$. $\mathcal{K}_W$ is the Wavelet kernel operator, defined as

$$(\mathcal{K}_W(\phi) * (v_t))(x) = \mathcal{W}^{-1}(R_{\phi} \cdot \mathcal{W}(v_t))(x) \quad (2)$$

where $\mathcal{W}$ is wavelet transform, $\mathcal{W}^{-1}$ is inverse wavelet transform, and R_{ϕ} is a neural network-parameterized representation of the kernel k in wavelet space with some parameter $\phi \in \mathcal{R}^P$. b is a bias function and σ is a non-linear activation function.

2.3 Adaptive Spatial Gate

FNO and WNO learn complementary features. The FNO branch operates in the Fourier domain to capture global structures and long-range correlations, whereas the WNO branch uses localised wavelet representations, which excel at representing sharp gradients and multiscale local features. Consequently, no single operator is uniformly optimal across the entire spatial domain: smoothly varying regions and strongly discontinuous regions demand different representational strengths.

To exploit this complementary nature, we use the Adaptive Gated Fusion Mechanism (GFM) [6] that performs spatially adaptive mixing of FNO and WNO features. Let

$$V_{\text{FNO}}, V_{\text{WNO}} \in \mathbb{R}^{B \times H \times W \times C}$$

represent the feature maps produced by FNO and WNO branches, respectively. We first concatenate these features along the channel dimension and pass them through a two-layer convolutional gate with kernel size $k = 5$ and spatial padding, followed by a sigmoid activation, producing a per-channel gating map

$\alpha \in \mathbb{R}^{B \times H \times W \times C}$:

$$\alpha = \sigma \left(\text{Conv}_k \left(\text{GELU} \left(\text{Conv}_k ([V_{\text{FNO}}, V_{\text{WNO}}]) \right) \right) \right).$$

The kernel size is chosen so that the gate’s effective receptive field (a 9-pixel window for $k = 5$ stacked twice) exceeds the characteristic width of the sharp features in the data — approximately 5–10 grid points for Burgers shocks at $\nu = 10^{-3}$. A single 1×1 convolution, having no spatial context, cannot express the spatial routing pattern and collapses to a near-uniform gate; the spatial receptive field is what enables the gate to localise sharp features.

The gate α at spatial location induces a piece-wise trust map that encodes the relative importance of FNO versus WNO. The fused representation is then obtained via an element-wise combination,

$$V_{\text{fused}} = \alpha \odot V_{\text{FNO}} + (1 - \alpha) \odot V_{\text{WNO}},$$

where the gating is applied element-wise per channel, allowing different feature channels to route independently. In regions where the underlying solution is smooth and globally coherent, the model learns $\alpha \approx 1$, thereby prioritising the FNO branch and preserving large-scale consistency. Whereas, near the sharp interfaces, discontinuities or highly localised structures, the gate tends to shift to 0, increasing the contribution of WNO and attenuating global spectral effects analogous to the Gibbs phenomenon that arise in purely Fourier-based representations. The GFM provides a router between the two experts, enabling data-dependent choice of expert while yielding a more robust and expressive hybrid operator.

2.4 Entropy-Regularised Gating

Trained with a data-fidelity loss alone, the gate is prone to a “mean-field” stagnation in which it settles at the indecisive equilibrium $\alpha \approx \frac{1}{2}$ everywhere, so that both branches are averaged uniformly and the spatial routing of Section 2.3 never emerges. To break this symmetry, we add a Shannon-entropy penalty on the gate activations to the training objective,

$$L_{\text{total}} = L_{\text{data}} + \lambda_{\text{ent}} \mathbb{E}_x [H(\alpha(x))], \quad H(\alpha) = -\alpha \log \alpha - (1 - \alpha) \log(1 - \alpha), \quad (3)$$

with $\lambda_{\text{ent}} = 0.01$. The penalty drives the per-location gate towards a near-binary (Bernoulli) state, encouraging the model to commit to the locally more appropriate basis — the Fourier branch in smooth, globally coherent regions and the wavelet branch near sharp gradients and discontinuities — rather than averaging the two. Combined with a small random initialisation of the gate convolution (which displaces the gate from the exact $\alpha = \frac{1}{2}$ saddle point), this regulariser produces decisive, spatially varying routing.

3. Experiments

3.1 Dataset

We evaluate AW-FNO on the spatial super-resolution of two-dimensional forced Navier–Stokes turbulence. The data are vorticity snapshots $\omega(x, y)$ of the forced incompressible Navier–Stokes equations on the periodic torus $[0, 2\pi)^2$ at a viscosity corresponding to Reynolds number $\text{Re} \approx 10^3$, providing a turbulent field rich in multi-scale coherent structures. The high-resolution (HR) target is the 128×128 snapshot; the low-resolution (LR) input is obtained by $4\times$ average-pooling to 32×32 followed by bicubic upsampling back to 128×128 , so the network learns the LR→HR residual on a fixed grid. We use 8,000 snapshots for training and 2,000 for testing, with fields normalised to zero mean and unit variance using statistics computed on the training set.

3.2 Baselines and Ablation Variants

We compare AW-FNO against the following models, all trained under an identical optimisation protocol (Section 3.4):

- **Bicubic** — the no-model baseline; the relative error of the bicubic-upsampled LR input itself, quantifying the difficulty of the task.
- **FNO** [4] — a standard Fourier Neural Operator with 4 layers, 32 hidden channels and 20×20 retained Fourier modes.
- **FNO-fat** — a parameter-augmented FNO with 128 hidden channels. This isolates whether additional spectral capacity alone closes the gap to AW-FNO, addressing the fairness of the comparison.
- **WNO** [5] — a Wavelet Neural Operator branch (4 layers, 32 channels, db4 wavelet, decomposition level 3).

To isolate the contribution of the adaptive gate, we evaluate three ablations of AW-FNO that share the same dual-branch backbone (FNO modes 20×20 , WNO db6 at level 3, 32 channels) and differ only in the fusion mechanism:

- **Fixed** $\alpha = 0.5$ — the two branches are averaged with a constant, non-learnable weight, removing all spatial routing.
- **1×1 gate** — the gate is a single 1×1 convolution followed by a sigmoid, i.e. a per-pixel channel mixing with *no* spatial receptive field.
- **Rich gate** ($k = 5$, 2 layers) — the proposed gate of Section 2.3, with a 9×9 effective receptive field, trained with the entropy regulariser ($\lambda_{\text{ent}} = 0.01$) and random gate initialisation.

3.3 Evaluation Metrics

We report (i) the *relative L_2 error*, our primary accuracy metric; (ii) the *relative H_1 error*, which is sensitive to the accuracy of spatial gradients and hence to sharp features; (iii) the *high-frequency spectral L_2 error*, the relative error restricted to wavenumbers above the LR Nyquist cutoff, which directly measures recovery of the small-scale content that bicubic upsampling discards and that Fourier methods tend to corrupt with Gibbs oscillations; and (iv) the *enstrophy error* $|\mathcal{E}_{\text{pred}} - \mathcal{E}_{\text{true}}|/\mathcal{E}_{\text{true}}$ with $\mathcal{E} = \frac{1}{2} \int \omega^2 dx$, a physically meaningful diagnostic of the predicted vorticity field. To characterise the gate we additionally report its mean Shannon entropy $\mathbb{E}_x[H(\alpha)]$ (lower means more decisive routing) and the Pearson correlation $\rho((1 - \alpha), |\nabla \omega|)$ between the WNO-routing weight and the vorticity-gradient magnitude of the target (positive means the gate selects the wavelet branch at sharp features).

3.4 Implementation Details

All models are trained for 500 epochs with the Adam optimiser (learning rate 10^{-3} , weight decay 10^{-4}), a step-decay schedule that halves the learning rate every 100 epochs, gradient clipping at 1.0, and a batch size of 16. Training minimises the relative L_2 loss; for the rich-gate model the entropy penalty of Eq. (3) is added with $\lambda_{\text{ent}} = 0.01$. All runs use a fixed random seed and are executed on a single 48 GB NVIDIA GPU. The checkpoint with the lowest test relative L_2 is retained for evaluation.

4. Results and Discussion

4.1 Super-Resolution Accuracy

Table 1 reports the test-set performance of all models on the 128^2 super-resolution task. Every neural operator improves on the bicubic baseline by more than an order of magnitude in relative L_2 error (from

Table 1: Super-resolution results on forced Navier–Stokes 128^2 (best-epoch test metrics; lower is better for all columns except where noted). High- f L_2 is the relative spectral error above the LR Nyquist cutoff. Gate entropy H is reported only for gated variants.

Model	Params	Rel L_2	Rel H_1	High- f L_2	Enstrophy	Gate H
Bicubic (no model)	—	0.4134	—	—	—	—
FNO	0.91M	0.0193	0.152	303.8	6.2e-4	—
FNO-fat ($c=128$)	14.6M	0.0198	0.157	325.3	5.6e-4	—
WNO	81.6M	0.0179	0.159	180.6	9.8e-4	—
AW-FNO (fixed $\alpha=0.5$)	89.3M	0.0210	0.194	265.7	1.3e-3	—
AW-FNO (1×1 gate)	89.3M	0.0134	0.116	173.7	1.4e-3	0.667
AW-FNO (rich gate)	89.3M	0.0094[†]	0.083	113.0	7.4e-4	0.017

[†]Rich-gate run still completing (epoch 317/500, error still decreasing); value provisional.

0.413 down to 0.019 for the plain operators and to 0.0094 for AW-FNO with the rich gate), confirming that the small-scale content removed by $4\times$ downsampling is genuinely recoverable and that the task is non-trivial.

Several findings emerge. (i) Among single operators the wavelet branch is strongest on the small scales: the WNO attains the lowest high-frequency spectral error (180.6 versus 303.8 for the FNO), consistent with its multi-resolution basis being better suited to localised vorticity gradients and with the Gibbs-suppression argument of Section 2.3. (ii) Additional Fourier capacity alone does not help: the parameter-augmented FNO-fat (14.6M parameters) is statistically indistinguishable from the standard FNO (0.0198 versus 0.0193), confirming that the gains below come from the dual-branch fusion rather than from raw model size. (iii) The fusion must be *learned*: a non-adaptive average of the two branches ($\alpha = 0.5$) is in fact worse than either branch alone (0.0210), whereas making the mixing coefficient learnable — even with a spatially-blind 1×1 gate — lowers the error to 0.0134, below both single operators. (iv) Accuracy and interpretability are, however, distinct: the 1×1 gate’s entropy stays near the indecisive value ($H \approx 0.67$, cf. $\ln 2 \approx 0.693$) and its routing is spatially flat (Section 4.3), so its gain stems from a learned per-channel re-weighting rather than from feature-aware spatial routing. Endowing the gate with a spatial receptive field (the rich gate) delivers both: it attains the best accuracy on every error metric (relative $L_2 \approx 0.0094$, a $\sim 48\%$ reduction over the best single operator and $\sim 30\%$ over the 1×1 gate) *and* a decisive, spatially structured gate ($H \approx 0.017$), which we examine in Sections 4.2–4.3.

4.2 Energy Spectrum

To assess the physical fidelity of the reconstructed fields beyond point-wise error, we compute the radially averaged kinetic-energy spectrum $E(k)$ and compare each model against the ground-truth HR field (Figure ??, generated by `scripts/energy_spectrum_sr.py`). The bicubic baseline collapses above the LR Nyquist wavenumber, having no energy to place there. The key diagnostic is the high- k inertial range: an operator that over-smooths loses energy there, while one that produces Gibbs ringing overshoots it.

4.3 Gate Routing Analysis

The central architectural claim of this work is that a spatially-resolved gate learns to route the wavelet branch to sharp features and the Fourier branch to smooth regions. We test this on the trained rich-gate model by extracting the per-block gate maps $\alpha(x, y)$ and correlating the WNO-routing weight $(1 - \alpha)$ with the vorticity-gradient magnitude $|\nabla\omega|$ of the target (Figure ??, produced by `scripts/analyze_gate_sr.py`). A positive correlation $\rho((1 - \alpha), |\nabla\omega|)$ is direct evidence of the claimed routing behaviour.

As a negative control, the 1×1 -gate variant produces an almost spatially constant gate: across the deepest fusion block its map has mean $\bar{\alpha} \approx 0.52$ but a spatial standard deviation of only $\approx 10^{-3}$, and a routing

correlation of merely $\rho \approx +0.03$. The gate therefore carries essentially no spatial information and reduces to a near-uniform average of the two branches, consistent with its collapsed entropy ($H \approx 0.68$, Table 1) and with the Phase-2 finding on the Burgers equation. This confirms that a spatial receptive field — not merely a learnable mixing coefficient — is what enables feature-aware routing.

5. Conclusions

This work proposes AW-FNO, an adaptive hybrid neural operator that combines the complementary strengths of Fourier and Wavelet representations for fluid flow super-resolution. With the integration of a parallel architecture with an adaptive gated fusion mechanism, we are able to capture the localised multiscale features. The entropy-regularised gating strategy promotes decisive, spatially varying selection between the two operators, thereby enhancing robustness near sharp gradients and discontinuities, and reducing spectral artefacts, such as Gibbs oscillations.

The AW-FNO introduces a hybrid neural operator framework that synergistically integrates the global spectral efficiency of FNOs with the multi-resolution localisation of WNOs for super-resolving turbulent flows. The fusion of parallel kernel architecture and adaptive GFM enables spatially varying feature selection. The model learns to prioritise FNO for smooth, large-scale coherent structures and WNO for sharp gradients, discontinuities and small-scale eddies. This routing behaviour requires the gate to have sufficient representational capacity: a single 1×1 convolution cannot express the spatial routing pattern, whereas a gate with a spatial receptive field can. This architectural design overcomes FNO’s limitations such as Gibbs oscillations and spectral bias near local discontinuities, resulting in high fidelity reconstruction, enhanced generalisation across unseen resolutions and reduced computational overhead compared to traditional DNS or standalone NOs.

Experiments demonstrate AW-FNO’s robustness on high-Reynolds turbulent benchmarks, achieving zero-shot super-resolution. Future extensions could be dealing with 3D unstructured meshes via graph integrations, further bridging the gap toward real-time CFD surrogates.

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